

MOHAMMED NAFEES H

Robotics & AI Engineering Student | ROS 2 | Robotics Controls | Raspberry Pi | Pi Camera | OpenCV

9629461267 | mohammednafees1007@gmail.com | [Kumbakonam, Tamil Nadu, India](#) | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Robotics & AI Engineering student at SASTRA University with hands-on experience from schooling through college in building robotics and embedded systems projects. Strong practical background in ROS 2, robotics controls, Raspberry Pi, Pi Camera, Ubuntu, OpenCV, and robot integration workflows. Earlier exposure includes Arduino-based prototyping during schooling. Interested in industrial robotics, autonomous systems, perception, navigation, and real-world deployment-oriented engineering.

EDUCATION

SASTRA Deemed University | B.Tech, Robotics & Artificial Intelligence
Current Student | Class of 2028 | Tamil Nadu, India

TECHNICAL SKILLS

- Robotics: ROS 2, robotics controls, robot integration, mobile robots, quadruped kinematics, sensor interfacing, control basics
- Programming: Python, C/C++, Linux shell scripting, Arduino basics
- Hardware: Raspberry Pi 5, Raspberry Pi Zero 2 W, Pi Camera, motor drivers, encoders, line sensors, sensor/actuator interfacing
- Software & Tools: Ubuntu 24.04, OpenCV, Git, VS Code, Coppeliasim, Fusion 360, EasyEDA/KiCad basics
- Interests: Industrial robotics, autonomous navigation, robot perception, embedded AI, reinforcement learning

EXPERIENCE

Confidential Industrial Robotics Project | ROS 2, Robotics Controls, System Integration

Project details withheld due to signed confidentiality / NDA obligations

- Contributing to an industry-level robotics project involving ROS 2-based workflows, controls-oriented development, embedded integration, and system-level coordination.
- Supporting real-world engineering tasks across robotics software, hardware coordination, and practical deployment-focused development in a professional setting.
- Responsible for sensor and actuator identification, evaluation, and procurement coordination for a project hardware budget of approximately Rs. 10 lakh.
- Working with industrial-grade components and practical engineering constraints while maintaining strict confidentiality.

PROJECTS

Hadi AI Hajatron | Arduino, Python, Tkinter, IR Sensing, A* Path Planning

- Built a grid-navigation robot system with live mapping, frontier exploration, IR obstacle sensing, encoder-based movement, saved map output, and A* route planning.

FruitVision AI - Fruit Quality Detector for SMS | FastAPI, YOLOv8x, Keras, CLIP, Next.js

- Built a quality-first fruit grading website that analyzes upload or camera images, predicts Fresh/Adulterant/Rotten quality, maps A/B/C grades, and displays annotated results.

AI Image Analyzer | Python, YOLOv8, OpenCV, BLIP, Transformers

- Built the backend image-analysis pipeline for a larger Visage project, combining object detection, privacy-safe face tokens, BLIP scene captions, and clean JSON output for app integration.

Quadruped Robot | Kinematics, controls, embedded integration

- Built inverse-kinematics motion logic for stable leg movement and practical actuation.

Schooling-Era Robotics Projects | Arduino-based prototyping, sensors, motor drivers

- Completed numerous small robotics and embedded builds during schooling; Arduino was used mainly in this phase for early prototyping, control experiments, wiring, and debugging.

LEADERSHIP & INVOLVEMENT

- Member, Robotics Club - active in robotics learning, project building, experimentation, and applied technical development.

EXPERIENCE HIGHLIGHTS

- Strong practical learning style built around reverse-solving problems, debugging from failure states, and improving through implementation.
- Built and tested robotics systems across software, electronics, and mechanical integration instead of working only in simulation.
- Comfortable with Linux-based robotics environments, hardware bring-up, sensor debugging, rapid prototyping, and project execution under confidentiality constraints.